1. Introduction & Integration

What is flexible bioelectronics

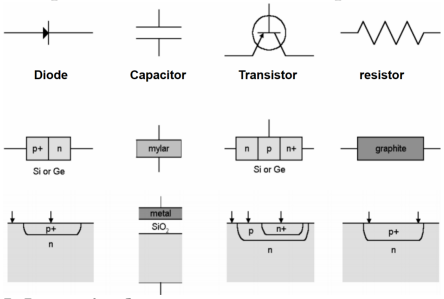
- **Electronics:** physics, engineering, technology and applications dealing with the emission, flow and control of electrons in vacuum and matter.
- **Flexible electronics:** the production of electronic components fabricated on a flexible substrate (typically polymer), giving thinner, lighter, softer, conformal and potentially stretchable systems.
- Main course goal: integrate high-performance electronics with soft, dynamic and biologically compatible interfaces for on-patient monitoring, diagnosis, therapy and human–machine interaction.

Electronic categories

- Digital, analogue, microelectronics, integrated circuits (ICs), power electronics, optoelectronics, semiconductor devices and embedded systems.
- **Discrete circuit:** individually packaged components; customizable but bulky and costly at high volume.
- **Monolithic circuit:** devices fabricated on one substrate; high density, low unit cost at volume, but requires cleanroom processing and is fixed after design.

Integration and Moore's law

- Integration evolved from vacuum tubes to transistors, ICs and nanoscale CMOS; resistor, capacitor, diode and transistor can be made in compatible semiconductor processes.



- **Moore's law:** transistor count in a dense IC approximately doubles every two years (an observation/trend, not a physical law).
- Scaling improves switching speed, energy use and miniaturization, but introduces purity, heat, fabrication and mass-production challenges.

Why flexible bioelectronics

- **Manufacturing cost:** Modern intel fabrication capital cost > 3,000,000,000.00 USD, while modest flex electronics fabrication capital cost < 3,000,000.00 USD
- **Market:** enables embedding simple intelligence with cost-efficient manufacturing techniques
- Conventional devices are hard/planar; organs and skin are soft, curved, moving and hydrated. Rigid interfaces cause discomfort, poor contact, motion artefact and low-fidelity signals.
- **Healthcare challenges:** centralized, slow, invasive, expensive, expertise-dependent testing and delayed diagnosis.
- **Desired properties:** conformality, flexibility/stretchability, stable adhesion, breathability, biocompatibility, low power and wireless communication.
- **Key design principle:** preserve high-performance active materials while reducing effective stiffness by thinning, patterning or embedding them in soft substrates.

Cross of biology and devices

- Biology | Interface | Device
- Interface needs: High performance, flexible, high though-put

E.g. Bio-Compatible Soft ECoG

- **Currently:** with wires, monitored by bulky medical equipment
- **Future:** wireless, monitored by portable interfaces, such as smartphones

E.g. Complex 3D Structures manufacturing

- **Method:** Stress-Induced Folding, Two-Photon Lithography, Direct-Writing Printing, 3D Photopatterning
- Can we exploit today' s the most advanced manufacturing technologies in semiconductor industry?
  -Yes, but selectively. Today' s advanced semiconductor manufacturing —EUV lithography, atomic-layer deposition, nanoscale etching, and heterogeneous integration—can be exploited, but access is limited by cost, equipment, materials, yield, and specialized expertise.
- Can we nano/microfab electronics from the existing highest-performance electronic materials and then transform them into 3D devices?
  -Yes, in principle. High-performance materials can be nano/microfabricated and integrated into 3D devices using stacking, transfer printing, folding, origami/kirigami, molding, and additive fabrication. However, preserving electrical performance during deformation, ensuring reliable interconnects, heat dissipation, and scalable manufacturing remain major challenges.

2. Devices & Materials

Band theory

- Atomic orbitals broaden into energy bands: valence band (VB), conduction band (CB), separated by band gap E_g.
- **Fermi level**: energy with maximum electron occupation probability at absolute zero.
- Metals: overlapping bands or $E_g \approx 0$; semiconductors: moderate gap; insulators: large gap. Examples: Si E_g = 1.12 eV, Ge = 0.66 eV, GaAs = 1.41 eV, $SiO_2 \approx 9$ eV, diamond = 5.47 eV.
- Donors (N, P, As) produce **n-type**; acceptors (B, Al, Ga) produce **p-type** material.

Electronic building blocks

- **Resistor/capacitor**: circuit elements and pressure, strain, temperature, gas and photodetectors.
- **Diode**: rectification; also LED, photodetector and solar cell. Emission color is set by semiconductor band gap.
- **Transistor**: switch, amplifier, sensor and ion-to-electron converter; important for amplifying weak biological signals.

$$\mu_{\text{lin}} = \frac{L}{WC_iV_D}\frac{\partial I_D}{\partial V_G}\ (V_D \ll V_G), \qquad \mu_{\text{sat}} = \frac{2}{W}$$

Crystal and material classes

- Si has diamond lattice; common structures include FCC, BCC and HCP. Miller notation: $[abc]$ direction, (abc) plane; crystal orientation affects wafer processing and anisotropic etching.
- Single crystal: long-range order and high electronic performance. Polycrystal: grain boundaries affect transport. Amorphous materials have lower thermal conductivity, can be transparent and stable, but deform more.
- Candidate materials: Si, III–V (GaAs), metal oxides, polymers, small molecules, graphene and carbon nanotubes.

Soft and emerging materials

- Conventional electronics fail at only about 1–3% strain; skin reaches $\sim$ 30% and joints can reach $\sim$ 100%.
- Soft substrates: PDMS, polyurethane, hydrogels and composites; matching Young's modulus to tissue reduces mismatch and irritation.
- **Liquid metals**: Ga, EGaIn (75.2% Ga/24.8% In) and Galinstan. They remain conductive while deforming (up to about 700% strain). Ga alloys form an oxide skin in oxygen; injection/vacuum filling patterns them into microchannels.
- **2D materials**: graphene, black phosphorus and MXene. Single/few atomic layers give high surface area, conductivity, transparency, flexibility and easy functionalization.

3. Mechanics & Design

Conformability

- Conformality is the ability to maintain intimate contact with rough, curved skin. Thickness dominates: thinner layers bend more easily and follow topography better.

$$D = \frac{Et^3}{12(1-\nu^2)}$$

- D: bending stiffness; E: Young's modulus; t: thickness; ν: Poisson ratio. The cubic t^3 dependence means a small thickness reduction greatly lowers stiffness.
- Engineer low E and low t, while retaining electrical and mechanical reliability.

Stretchable layouts

- **Wavy/wrinkled**: prestrain a soft substrate, attach a stiff thin film, then release; out-of-plane waves accommodate extension.
- **Serpentine**: meandering ribbons rotate, bend and buckle so material strain is much lower than applied strain. Important parameters: width w, arc radius R, arc angle α, arm length l. Smaller w/R and longer arms generally improve stretchability.
- **Island–bridge**: rigid functional islands connected by soft, wavy or serpentine bridges. Non-coplanar layouts are usually more stretchable; fractal interconnects increase coverage and compliance.
- Origami/kirigami and composite structures add geometric deformation, fatigue resistance and mechanical buffering.

Interface requirements

- **Adhesion**: must be strong, conformal and durable under motion, sweat and repeated loading. Dry adhesion uses van der Waals, electrostatic, friction or mechanical interlocking; wet adhesion uses capillary, hydrogen-bonding, viscous and chemical interactions.
- **Breathability**: permits air/moisture exchange and reduces sweat accumulation. **Biocompatibility**: no toxicity, irritation, allergy or disruption of tissue function. **Biodegradability**: safe dissolution can avoid retrieval surgery and persistent waste.

4. Biomarkers & Wearable Sensing

Biofluids

Fluid	Advantages	Main limitations
Sweat	Non-invasive; surface-accessible; electrolytes/metabolites/hormones	low volume, evaporation, contamination, rate and site variations
ISF	Close to blood for small molecules; rich biomarkers	invasive sampling, low volume, discomfort
Tears	Continuous, non-invasive secretion	very low volume; eye engineering; weak blood correlation
Saliva	Non-invasive, continuously available, large daily volume	viscosity; food/drink contamination; production varies
Urine	Rich and abundant biomarker source	wearable use limited to urination events

Biomarker classes

- **Electrolytes**: Na^+, K^+, Mg^{2+}, Ca^{2+}, Cl^-; hydration, cardiac, liver and metabolic status.
- **Metabolites**: glucose, lactate, uric acid, alcohol and amino acids; diabetes, metabolism and exercise. Sweat glucose is typically 10–100 times lower than blood glucose.
- **Hormones**: cortisol, estradiol and peptides; stress, endocrine, reproductive, sleep and mood monitoring.
- Proteins/enzymes (cytokines, troponin, LDH), nucleic acids, bacteria/cells, drugs and other small molecules support cardiovascular, cancer, wound, infection and therapeutic monitoring.

Recognition and transduction

- Biorecognition elements: enzymes, antibodies/affinity proteins, peptides, aptamers, receptors and molecularly imprinted polymers (MIPs). They bind/catalyse the target and are immobilized on a transducer.
- Enzyme example: glucose oxidase catalyses glucose, producing H_2O_2; electrode redox current increases with analyte concentration.
- Aptamer: synthetic ssDNA/RNA binds target; target-induced conformational change changes current, voltage, impedance, fluorescence or color. Modes include sandwich, dissociation, competitive replacement and structure switching.

Method	Principle / typical use
Amperometric	redox current $\propto$ concentration; glucose/lactate
Potentiometric	equilibrium potential; ion-selective sensing
Voltammetric	current versus applied potential; electroactive species
Impedimetric	binding changes AC impedance
Optical	absorption, reflection, fluorescence or color
Electromechanical	pressure, strain, bending, acceleration or vibration

5. Processing I: Cleanroom & Lithography

Safety and contamination

- Identify chemical, electrical, fire and sharps hazards; use PPE, correct containers, exits/showers and MSDS. HF is a contact poison and attacks glass; never treat it as an ordinary acid.
- Cleanroom defects include particles, organics, ions, moisture and surface damage. DI water is ion-free: at 25°C, $[H^+][OH^-] = 10^{-14}$, $[H^+] = [OH^-] = 10^{-7}$ and resistivity $\approx$ 18.5 M cm.

Wafer cleaning

1. DI-water ultrasonic/megasonic clean removes particles and soluble ions.
2. RCA SC-1: DI:H_4NOH:H_2O_2 = 5 : 1 : 1, typically 75–80°C for 10 min; organic/particle clean, forms $\sim$ 1 nm SiO_2.
3. Optional HF 1:50–1:100, about 15 s at room temperature, strips oxide.
4. RCA SC-2: DI:HCl:H_2O_2 = 6 : 1 : 1, 75–80°C for 10 min; removes ionic/metal contaminants and passivates the surface.

- Alternatives: piranha ($H_2SO_4 : H_2O_2 = 3 : 1$; strongly exothermic), ozone, solvent, UV-ozone and plasma cleaning.

Photolithography

- Typical flow: substrate clean $\rightarrow$ photoresist coat/spin $\rightarrow$ soft bake $\rightarrow$ mask align/expose $\rightarrow$ develop $\rightarrow$ hard bake $\rightarrow$ etch/deposit/lift-off.
- Positive resist: exposed region dissolves in developer; negative resist: exposed region cross-links and remains.
- Photomask: patterned Cr on flat quartz/film. Alignment marks and backside/IR alignment enable multilayer MEMS processing.
- Yellow light ($\sim$ 590–620 nm) suppresses UV exposure of resist. Dose is optical energy per area; under/overexposure changes linewidth and profile. Smaller wavelength improves resolution.
- Phase-shift mask changes adjacent phase by 180° for destructive interference; optical proximity correction predistorts the mask.

Other patterning

- **E-beam lithography**: direct-write, high resolution, slow and serial; useful for research and masks.
- **Soft lithography**: elastomeric stamp/mold; low cost and compatible with large-area flexible substrates.
- **Laser direct writing/engraving**: maskless, rapid prototyping and direct formation/patterning of conductive features.

6. Processing II: Deposition & Etching

Vacuum and PVD

- 1 atm = 760 torr = 1.013×10^5 Pa; 1 torr = 133.3 Pa. Vacuum reduces collisions and contamination. Base pressure is reached before process gas; sputtering then introduces Ar at working pressure (typically millitorr range) to sustain plasma.
- **Evaporation**: source is heated (thermal/e-beam); atoms travel ballistically to substrate, giving line-of-sight coverage.
- **Sputtering**: Ar plasma ions bombard target and eject atoms; high-energy transport, high rate and better suitability for refractory metals such as Ti/W. Co-sputtering controls composition.
- Interconnect trade-off: Al is cheap, stable, easy to deposit/etch and diffuses little in Si/SiO_2; Cu is more conductive and reduces RC delay below about 130 nm, but requires diffusion barriers/processing control.

CVD, ALD and solution deposition

- **CVD**: gaseous precursors react/decompose at the surface; controls include temperature, pressure, gas type/flow and time, determining thickness, crystallinity and film quality.

$$SiH_4(g) \rightarrow Si(s) + 2H_2(g)$$

- PECVD plasma increases ion energy, enabling higher rate at lower temperature. **ALD** is self-limiting, layer-by-layer, highly uniform and gives precise thickness control.
- Solution methods: spin, spray, chemical bath, dip, blade, drop and slot-die coating. They are inexpensive and flexible-substrate compatible, but uniformity, morphology and electrical performance depend strongly on process parameters.

Etching

- **Etching** removes desired material selectively; **corrosion** is unwanted attack. Wet etching uses liquid, dry etching uses gas/plasma. Important metrics: etch rate, selectivity and anisotropy.
- Isotropic etch attacks laterally and vertically; anisotropic etch preserves sidewalls and directionality.
- Examples: HF etches SiO_2; $FeCl_3$ etches Cu; KI/I_2 etches Au. KOH etches Si anisotropically with approximate rate ratio $(111) : (110) : (100) = 1 : 600 : 400$ and can use SiO_2/Si_3N_4 masks.

Exam checkpoints

- Always connect **material → property → process → device function**. Example: thin stiff Si patterned into a serpentine on PDMS gives high electronics performance with low effective stiffness.
- For a wearable biosensor, specify biofluid, target biomarker, recognition chemistry, transduction mode, sampling constraint and mechanical/biocompatibility requirement.
- For fabrication questions, distinguish substrate cleaning, pattern definition, film deposition and selective removal; state why a chosen method fits the material and geometry.